

# Real Analysis I — Homework 6

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Unless otherwise specified, all functions are defined on  $E \subset \mathbb{R}^n$  where  $E \in \mathcal{M}$ .

1. If  $\nu$  is a signed measure, then  $E$  is  $\nu$ -null iff  $|\nu|(E) = 0$ . Also, if  $\nu$  and  $\mu$  are signed measures,  $\nu \perp \mu$  iff  $|\nu| \perp \mu$  iff  $\nu^+ \perp \mu$  and  $\nu^- \perp \mu$ .
2. Suppose  $\{\nu_j\}$  is a sequence of positive measures. If  $\nu_j \perp \mu$  for all  $j$ , then  $(\sum_{j=1}^{\infty} \nu_j) \perp \mu$ ; and if  $\nu_j \ll \mu$  for all  $j$ , then  $(\sum_{j=1}^{\infty} \nu_j) \ll \mu$ .
3. Suppose that  $\mu, \nu$  are positive measures on  $(X, \mathcal{M})$  with  $\nu \ll \mu$ , and let  $\lambda = \mu + \nu$ . If  $f = \frac{d\nu}{d\lambda}$ , then  $0 \leq f < 1$   $\mu$ -a.e., and  $\frac{d\nu}{d\mu} = \frac{f}{1-f}$ .
4. Suppose  $f \in \text{BV}([a, b])$ , and  $f(x) \geq c > 0$  on  $[a, b]$ . Show that  $\frac{1}{f} \in \text{BV}([a, b])$ .
5. Show that  $f \in \text{BV}([a, b])$  iff there exists a non-decreasing function  $\phi : [a, b] \rightarrow \mathbb{R}$  such that

$$|f(y) - f(x)| \leq \phi(y) - \phi(x), \quad \text{for all } a \leq x < y \leq b.$$

6. Suppose  $f \in \text{AC}([a, b])$ . Show that

$$\int_a^b |f'(x)| \, dx = \text{TV}(f, [a, b]).$$

Hint: We showed  $\geq$  in class. For  $\leq$ , it is easy to show that  $\int_a^b s(x)f'(x) \, dx \leq \text{TV}(f)$  where  $s(x) = \sum_{i=1}^p c_i \chi_{[x_{i-1}, x_i)}(x)$  and  $|c_i| \leq 1$  for any partition  $\Delta : a = x_0 < x_1 < \dots < x_n = b$ . Let  $\phi_k$  be a sequence of simple functions such that  $\phi_k \rightarrow f'$ , and  $s_k(x)$  be the sign of  $\phi_k(x)$ . Apply DCT to  $s_k f'$ .

7. Suppose  $f \in L([a, b])$ . Show that for a.e.  $x \in [a, b]$ , there is

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_{x-h_1}^{x+h_2} |f(y) - f(x)| \, dy = 0,$$

where  $h_1, h_2 \geq 0$  and  $h = h_1 + h_2$ .

8. Suppose  $f \in L^2((0, \pi))$ . Show that the following two inequalities cannot hold simultaneously:

$$\int_0^\pi (f(x) - \sin x)^2 \, dx \leq \frac{4}{9}, \quad \int_0^\pi (f(x) - \cos x)^2 \, dx \leq \frac{1}{9}.$$

9. Suppose  $p, q, r > 0$  satisfy  $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$ . Then for any functions  $f \in L^p(E)$ ,  $g \in L^q(E)$  and  $h \in L^r(E)$ , there is

$$\int_E |fgh| \, dx \leq \|f\|_p \|g\|_q \|h\|_r.$$

10. Suppose  $f \in L^p(\mathbb{R})$  where  $1 < p < \infty$ , and  $q$  is the conjugate of  $p$ . Let  $F(x) = \int_a^x f(t) \, dt$ . Show that  $F(x+h) - F(x) = o(|h|^{1/q})$  as  $h \rightarrow 0$  for all  $x$ .